K.R. Mangalam University

Weather Data Visualizer – Assignment Report

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Course: B.Tech CSE (AI & ML)
Section: B
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Submitted as part of the course: Programming for Problem Solving using Python

1. Introduction

This report presents the complete Weather Data Visualizer project, developed as part of the Programming for Problem Solving using Python course. The purpose of this assignment is to analyze real-world weather data using Python-based tools such as pandas, numpy, and matplotlib. The project includes:

- Reading and cleaning the dataset
- Processing missing values
- Statistical analysis of temperature, humidity, and wind speed
- Generating visual plots
- Understanding real-world climate patter

Weather data analysis helps understand environmental behavior patterns and supports climate-based decision-making. This project demonstrates the application of Python in real data processing, numerical computation, and graphical visualization.

2. Dataset Description

The dataset used in this project is **DailyDelhiClimateTrain.csv**, which contains several years of daily weather records from Delhi. It includes columns such as:

- Date
- Mean Temperature
- Humidity
- Wind Speed
- Mean Atmospheric Pressure The dataset required preprocessing before analysis. This included date conversion, handling missing values using forward and backward fill techniques, and converting numeric fields to appropriate datatypes.

3. Methods & Algorithms Used

Pandas was used extensively for:

- Reading CSV files
- Cleaning missing values
- Filtering and selecting columns
- Grouping and resampling data

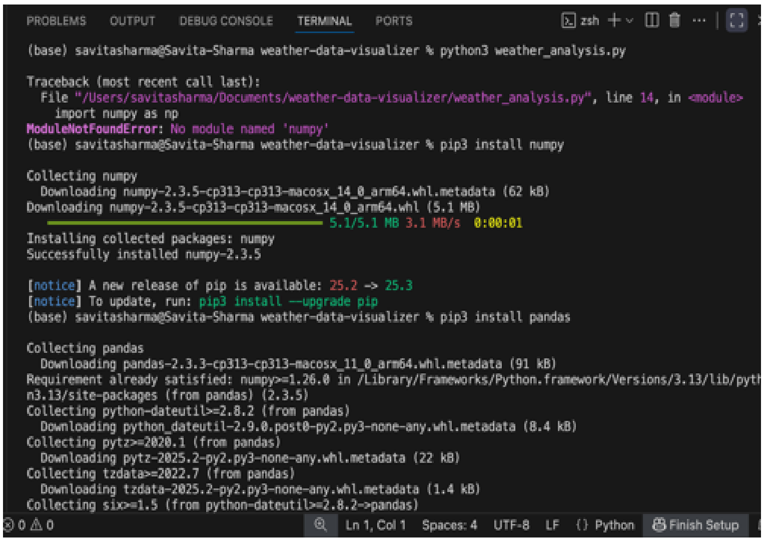
Numpy supported numerical computation such as:

- Mean
- Minimum
- Maximum
- Standard Deviation

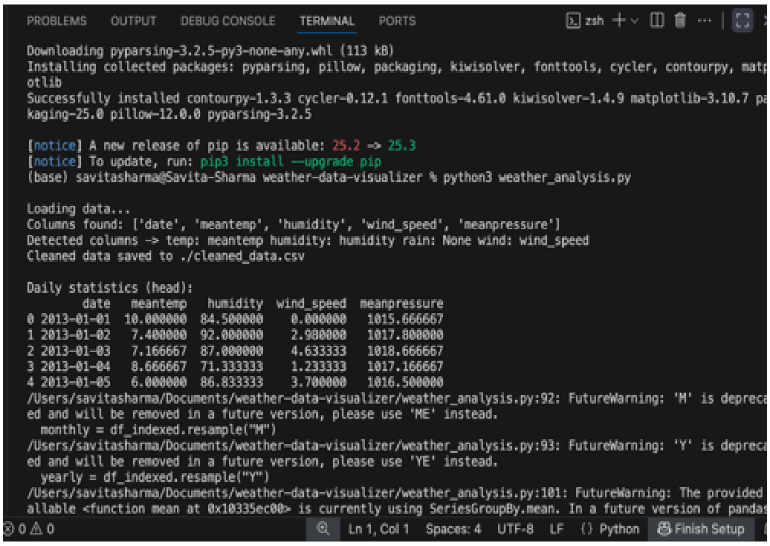
Matplotlib was used for:

- Line plots for temperature trends
- Scatter plots for humidity vs temperature
- Combined plots for multi-variable visualization

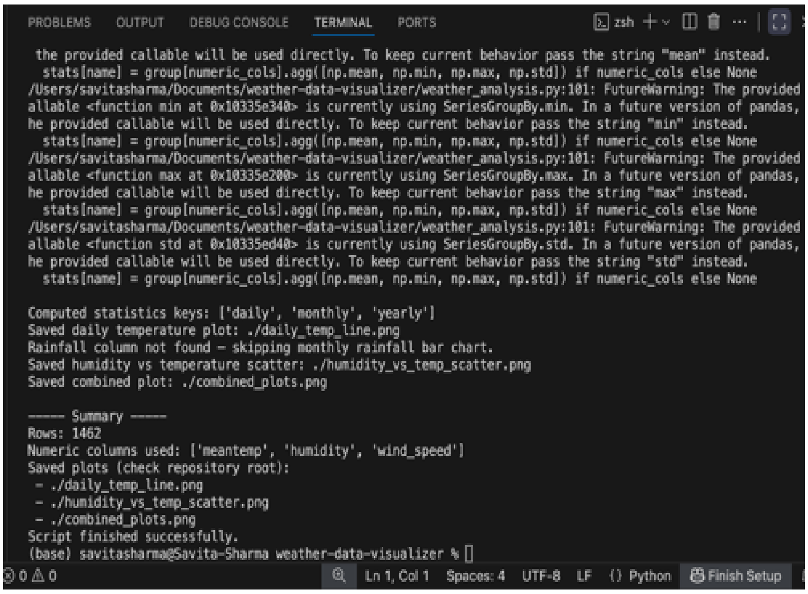
4. Code Execution Screenshots



```
(base) savitasharma@Savita-Sharma weather-data-visualizer % python3 weather_analysis.py

Traceback (most recent call last):
  File "/Users/savitasharma/Documents/weather-data-visualizer/weather_analysis.py", line 14, in <module>
    import numpy as np
ModuleNotFoundError: No module named 'numpy'
(base) savitasharma@Savita-Sharma weather-data-visualizer % pip3 install numpy

Collecting numpy
  Downloading numpy-2.3.5-cp313-cp313-macosx_14_0_arm64.whl.metadata (62 kB)
Downloading numpy-2.3.5-cp313-cp313-macosx_14_0_arm64.whl (5.1 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 5.1/5.1 MB 3.1 MB/s  0:00:01
Installing collected packages: numpy
Successfully installed numpy-2.3.5

[notice] A new release of pip is available: 25.2 -> 25.3
[notice] To update, run: pip3 install --upgrade pip
(base) savitasharma@Savita-Sharma weather-data-visualizer % pip3 install pandas

Collecting pandas
  Downloading pandas-2.3.3-cp313-cp313-macosx_11_0_arm64.whl.metadata (91 kB)
Requirement already satisfied: numpy>=1.26.0 in /Library/Frameworks/Python.framework/Versions/3.13/lib/pyth
n3.13/site-packages (from pandas) (2.3.5)
Collecting python-dateutil>=2.8.2 (from pandas)
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting pytz>=2020.1 (from pandas)
  Downloading pytz-2025.2-py2.py3-none-any.whl.metadata (22 kB)
Collecting tzdata>=2022.7 (from pandas)
  Downloading tzdata-2025.2-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas)
```

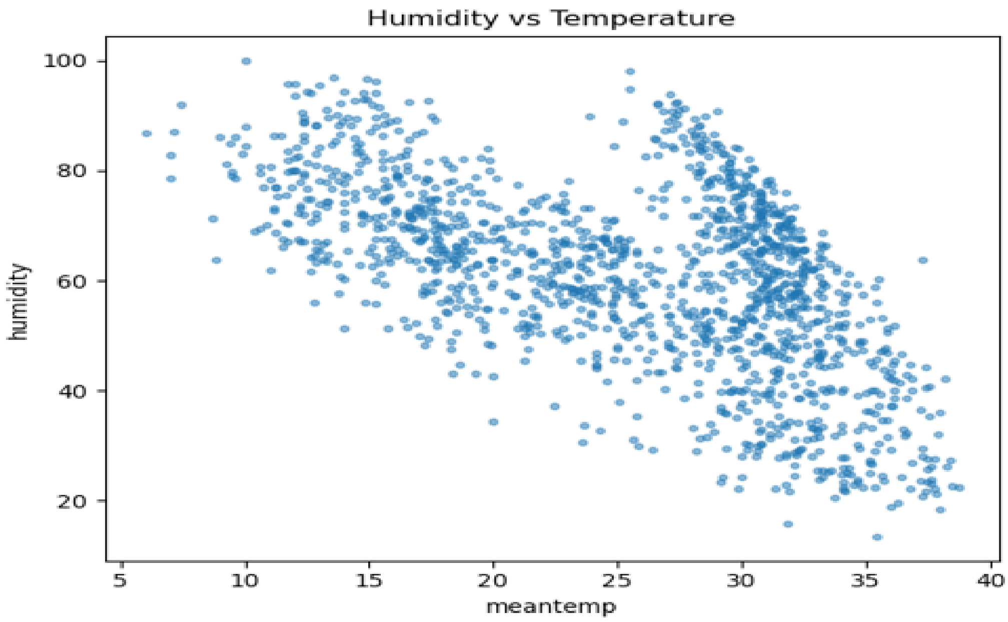


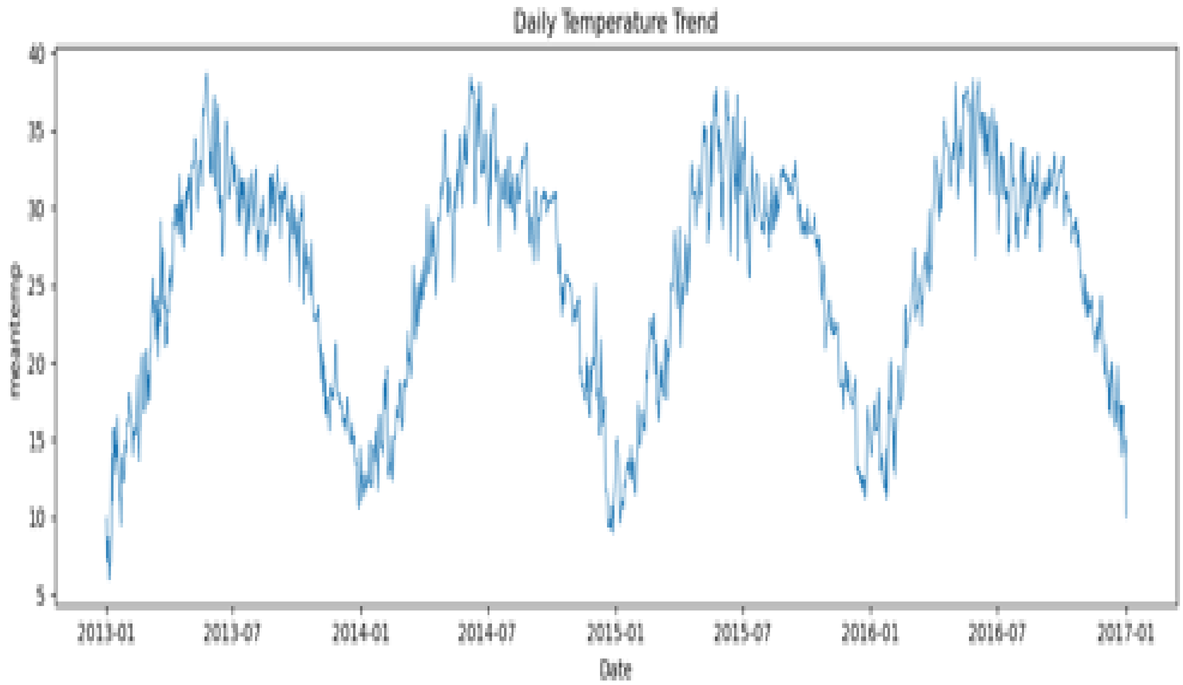
```
Downloading pyparsing-3.2.5-py3-none-any.whl (113 kB)
Installing collected packages: pyparsing, pillow, packaging, kiwisolver, fonttools, cycler, contourpy, matp
otlib
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.61.0 kiwisolver-1.4.9 matplotlib-3.10.7 pa
kaging-25.0 pillow-12.0.0 pyparsing-3.2.5

[notice] A new release of pip is available: 25.2 -> 25.3
[notice] To update, run: pip3 install --upgrade pip
(base) savitasharma@Savita-Sharma weather-data-visualizer % python3 weather_analysis.py

Loading data...
Columns found: ['date', 'meantemp', 'humidity', 'wind_speed', 'meanpressure']
Detected columns -> temp: meantemp humidity: humidity rain: None wind: wind_speed
Cleaned data saved to ./cleaned_data.csv

Daily statistics (head):
        date   meantemp   humidity  wind_speed  meanpressure
0 2013-01-01  10.000000  84.500000    0.000000   1015.666667
1 2013-01-02   7.400000  92.000000    2.980000   1017.800000
2 2013-01-03   7.166667  87.000000    4.633333   1018.666667
3 2013-01-04   8.666667  71.333333    1.233333   1017.166667
4 2013-01-05   6.000000  86.833333    3.700000   1016.500000
/Users/savitasharma/Documents/weather-data-visualizer/weather_analysis.py:92: FutureWarning: 'M' is depreca
ed and will be removed in a future version, please use 'ME' instead.
  monthly = df_indexed.resample("M")
/Users/savitasharma/Documents/weather-data-visualizer/weather_analysis.py:93: FutureWarning: 'Y' is depreca
ed and will be removed in a future version, please use 'YE' instead.
  yearly = df_indexed.resample("Y")
/Users/savitasharma/Documents/weather-data-visualizer/weather_analysis.py:101: FutureWarning: The provided
allable <function mean at 0x10335ec00> is currently using SeriesGroupBy.mean. In a future version of pandas
```

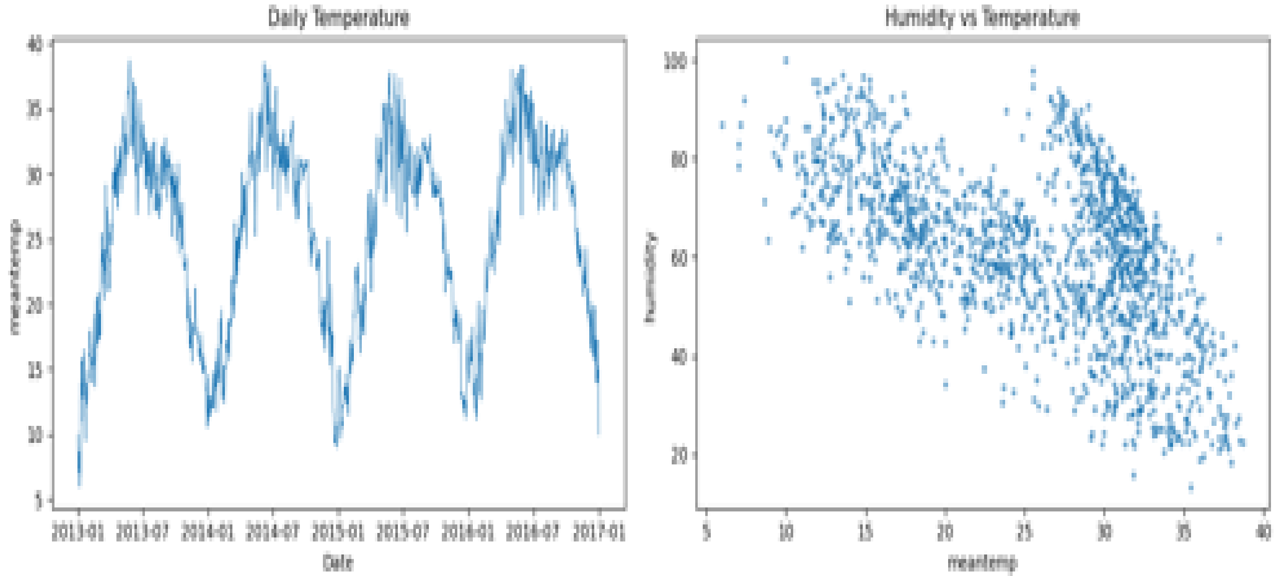


```
 the provided callable will be used directly. To keep current behavior pass the string "mean" instead.
  stats[name] = group[numeric_cols].agg([np.mean, np.min, np.max, np.std]) if numeric_cols else None
/Users/savitasharma/Documents/weather-data-visualizer/weather_analysis.py:101: FutureWarning: The provided
allable <function min at 0x10335e340> is currently using SeriesGroupBy.min. In a future version of pandas,
he provided callable will be used directly. To keep current behavior pass the string "min" instead.
  stats[name] = group[numeric_cols].agg([np.mean, np.min, np.max, np.std]) if numeric_cols else None
/Users/savitasharma/Documents/weather-data-visualizer/weather_analysis.py:101: FutureWarning: The provided
allable <function max at 0x10335e200> is currently using SeriesGroupBy.max. In a future version of pandas,
he provided callable will be used directly. To keep current behavior pass the string "max" instead.
  stats[name] = group[numeric_cols].agg([np.mean, np.min, np.max, np.std]) if numeric_cols else None
/Users/savitasharma/Documents/weather-data-visualizer/weather_analysis.py:101: FutureWarning: The provided
allable <function std at 0x10335ed40> is currently using SeriesGroupBy.std. In a future version of pandas,
he provided callable will be used directly. To keep current behavior pass the string "std" instead.
  stats[name] = group[numeric_cols].agg([np.mean, np.min, np.max, np.std]) if numeric_cols else None

Computed statistics keys: ['daily', 'monthly', 'yearly']
Saved daily temperature plot: ./daily_temp_line.png
Rainfall column not found - skipping monthly rainfall bar chart.
Saved humidity vs temperature scatter: ./humidity_vs_temp_scatter.png
Saved combined plot: ./combined_plots.png

----- Summary -----
Rows: 1462
Numeric columns used: ['meantemp', 'humidity', 'wind_speed']
Saved plots (check repository root):
 - ./daily_temp_line.png
 - ./humidity_vs_temp_scatter.png
 - ./combined_plots.png
Script finished successfully.
(base) savitasharma@Savita-Sharma weather-data-visualizer % []
```

5. Generated Visualizations







6. Observations

After processing and analyzing the dataset, the following observations were made:

Temperature Trends: Daily temperatures show periodic fluctuations representing seasonal cycles. Summer months show higher peaks while winter months show lower minimums.

Humidity Patterns: Humidity demonstrates an inverse relationship with temperature. Higher temperatures generally correlate with lower humidity levels.

Wind Speed Behavior: Wind speed fluctuates moderately and does not show a strong correlation with temperature or humidity.

Combined Visual Patterns: The multi-plot figure helps compare different weather parameters together. This supports better understanding of climate behavior across time.

7. Conclusion

This project successfully demonstrates Python-based data analysis, including data cleaning, statistical exploration, and visual representation. The generated plots provide insight into temperature behavior, humidity patterns, and wind speed variations across the dataset. Through this assignment, important skills were strengthened:

• Working with real-world datasets
• Using pandas for cleaning and manipulation
• Using numpy for statistical computation
• Using matplotlib for insightful plotting Overall, the Weather Data Visualizer project highlights how Python can be effectively used for environmental data processing and scientific analysis.